Deterministic Optimization

Large-Scale Optimization

Cutting Stock Problem

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Gilmore-Gomory Formulation

Large-Scale Optimization

Learning Objectives for this lesson

- Discover a linear programming formulation for the cutting-stock problem
- Understand the key feature of this formulation: the large number of variables

Cutting Stock Problem

- A paper company produces large rolls of paper
- Each roll has width $W = 100$
- Customer demand:
  - 150 rolls of width 25
  - 200 rolls of width 35
  - 300 rolls of width 45
- Company needs to meet the demand exactly, with minimum # of big rolls

Key Concept is Pattern

- We define a pattern to be a feasible way to cut a large roll into smaller rolls that customer demanded

Item	Patterns													Demand
w_i	1	2	3	4	5	6	7	8	9	10	11	....	N	b_i
25	4	3	2	2	2	1	1	1	1	0	0	....	a_1	150
35	0	0	0	1	0	2	0	1	0	1	0	....	a_2	200
45	0	0	0	0	1	0	1	0	0	0	1	....	a_3	300

- There are a finite number of patterns, but N can be very large

Decision Variables

- Decision variables:
    - x_i: the number of large rolls cut in pattern i for $i = 1,2,\ldots,N$

	x_1	x_2	x_3										x_N	
Item	Patterns													Demand
w_i	1	2	3	4	5	6	7	8	9	10	11	....	N	b_i
25	4	3	2	2	2	1	1	1	1	0	0	....	a_1	150
35	0	0	0	1	0	2	0	1	0	1	0	....	a_2	200
45	0	0	0	0	1	0	1	0	0	0	1	....	a_3	300

Constraints

- Meet the demand with equality

Item	Patterns													Demand
w_i	1	2	3	4	5	6	7	8	9	10	11	....	N	b_i
25	4	3	2	2	2	1	1	1	1	0	0	....	a_1	150
35	0	0	0	1	0	2	0	1	0	1	0	....	a_2	200
45	0	0	0	0	1	0	1	0	0	0	1	....	a_3	300

$$4x_1 + 3x_2 + 2x_3 + 2x_4 + \cdot + a_1 x_N = 150$$
$$0x_1 + 0x_2 + 0x_3 + 1x_4 + \cdot + a_2 x_N = 200$$
$$0x_1 + 0x_2 + 0x_3 + 0x_4 + \cdot + a_2 x_N = 250$$
$$x_1, x_2, \ldots, x_N \geq 0$$

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$
$$\mathbf{x} \geq \mathbf{0}$$

Objective Function

- The objective function:
  - Minimize the total number of large rolls used
  - $\min x_1 + x_2 + \cdots + x_N$

- The Gilmore-Gomory linear program formulation
  - $\min \sum_{i=1}^{N} x_i$
  - s.t. $Ax = b$
  - $\qquad x \geq 0$
  - where the columns of A are the patterns to cut one large roll
  - b is the amount of demand of each size of smaller rolls

Key Feature of GG Formulation

- The key feature of the Gilmore-Gomory formulation is the number of variables
    - The number of ways to cut a large roll into smaller ones is usually astronomical
    - Therefore, the number of variables is very large
    - This is the first time in this course that we encountered a linear program that is so large to even write down the complete model

Summary

- We learned:
  - Gilmore-Gomory formulation of cutting stock problem
  - Large-scale optimization with a large number of variables